

**BURGAS, Bulgaria**WMO: **156550**Lat: **42.570N**Lon: **27.515E**Elev: **135**StdP: **14.62**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 18.7       | 22.6 | 10.6                          | 9.5 | 20.8 | 15.5 | 12.1 | 24.8 | 40.4                  | 30.1 | 33.7 | 29.7 | 12.9                  | 0    | 0.550 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.5                   | 89.3            | 71.0 | 86.3 | 71.0 | 84.3 | 70.4 | 75.6                | 83.4 | 74.2 | 82.1 | 72.8 | 80.8 | 11.7                 | 120  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 73.4                            | 124.7 | 80.6 | 71.6 | 117.2 | 79.2 | 70.0 | 110.8 | 77.9 | 39.3          | 83.4 | 37.8 | 81.7 | 36.5 | 81.2 | 82.2              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |             |             | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |            |              |            |              |            |              |    |    |
|-------------------|-------------|-------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|------------|--------------|------------|--------------|------------|--------------|----|----|
|                   |             |             | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years |              | n=20 years |              | n=50 years |              |    |    |
| 1%                | 2.5%        | 5%          | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min        | Max          | Min        | Max          | Min        | Max          | DB | WB |
| (a)               | (b)         | (c)         | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)        | (l)          | (m)        | (n)          | (o)        | (p)          |    |    |
| (4) <b>31.5</b>   | <b>26.9</b> | <b>22.7</b> | <b>13.2</b>                | <b>95.9</b> | <b>4.0</b>         | <b>4.5</b> | <b>10.3</b>                                        | <b>99.1</b> | <b>8.0</b> | <b>101.8</b> | <b>5.8</b> | <b>104.3</b> | <b>2.9</b> | <b>107.5</b> |    |    |
| (5)               |             |             | <b>12.1</b>                | <b>78.9</b> | <b>4.0</b>         | <b>1.6</b> | <b>9.2</b>                                         | <b>80.0</b> | <b>6.9</b> | <b>80.9</b>  | <b>4.7</b> | <b>81.8</b>  | <b>1.8</b> | <b>82.9</b>  |    |    |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 55.9   | 37.0  | 39.9  | 45.5  | 52.6  | 62.0  | 70.3  | 74.9  | 74.8  | 67.0  | 57.6  | 48.9  | 39.7  |
|                                                                                  | DBStd       | 14.83  | 7.88  | 8.22  | 7.30  | 6.24  | 5.59  | 4.82  | 3.72  | 4.13  | 5.30  | 6.53  | 7.84  | 8.11  |
|                                                                                  | HDD50       | 1380   | 408   | 293   | 180   | 41    | 1     | 0     | 0     | 0     | 0     | 15    | 111   | 331   |
|                                                                                  | HDD65       | 4235   | 869   | 702   | 606   | 372   | 126   | 11    | 0     | 0     | 41    | 238   | 486   | 784   |
|                                                                                  | CDD50       | 3544   | 3     | 10    | 39    | 120   | 372   | 610   | 770   | 769   | 511   | 251   | 77    | 12    |
|                                                                                  | CDD65       | 925    | 0     | 0     | 0     | 1     | 31    | 171   | 306   | 304   | 101   | 10    | 1     | 0     |
|                                                                                  | CDH74       | 6999   | 0     | 0     | 2     | 24    | 192   | 1174  | 2487  | 2512  | 552   | 52    | 4     | 0     |
|                                                                                  | CDH80       | 1917   | 0     | 0     | 0     | 2     | 30    | 284   | 750   | 751   | 94    | 6     | 0     | 0     |
| Wind                                                                             | WSAvg       | 9.5    | 9.2   | 10.1  | 10.3  | 9.5   | 9.0   | 8.8   | 9.2   | 9.8   | 9.9   | 9.6   | 8.9   | 9.5   |
| Precipitation                                                                    | PrecAvg     | 23.1   | 1.5   | 1.8   | 1.5   | 2.0   | 1.8   | 2.0   | 1.5   | 1.0   | 1.8   | 2.2   | 2.5   | 2.1   |
|                                                                                  | PrecMax     | 30.0   | 3.7   | 4.4   | 3.5   | 3.8   | 4.4   | 5.2   | 4.4   | 3.4   | 6.3   | 7.9   | 7.4   | 4.4   |
|                                                                                  | PrecMin     | 14.6   | 0.0   | 0.2   | 0.2   | 0.2   | 0.3   | 0.1   | 0.0   | 0.0   | 0.0   | 0.1   | 0.6   | 0.2   |
|                                                                                  | PrecStd     | 4.0    | 1.1   | 1.1   | 0.9   | 0.8   | 1.0   | 1.3   | 1.2   | 0.8   | 1.5   | 1.7   | 1.5   | 1.2   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 60.5  | 66.0  | 71.8  | 77.3  | 83.8  | 90.0  | 93.5  | 93.4  | 86.9  | 79.2  | 73.4  | 64.2  |
|                                                                                  |             | MCWB   | 53.1  | 55.8  | 57.1  | 61.2  | 67.2  | 71.8  | 70.3  | 71.9  | 67.7  | 65.3  | 61.5  | 56.7  |
|                                                                                  | 2%          | DB     | 55.2  | 60.5  | 66.3  | 71.4  | 78.9  | 85.9  | 89.3  | 89.4  | 82.5  | 74.3  | 68.0  | 59.0  |
|                                                                                  |             | MCWB   | 49.3  | 52.6  | 54.0  | 58.7  | 65.7  | 70.5  | 70.7  | 71.5  | 68.4  | 63.9  | 60.0  | 53.6  |
|                                                                                  | 5%          | DB     | 50.3  | 55.8  | 61.3  | 66.4  | 75.4  | 82.8  | 86.2  | 86.3  | 79.2  | 70.6  | 63.3  | 54.6  |
|                                                                                  |             | MCWB   | 46.4  | 49.5  | 52.0  | 56.5  | 64.1  | 69.4  | 70.9  | 71.6  | 67.5  | 62.5  | 58.2  | 50.9  |
|                                                                                  | 10%         | DB     | 47.0  | 51.6  | 56.9  | 62.6  | 72.0  | 80.5  | 84.1  | 84.1  | 77.0  | 67.8  | 59.4  | 51.5  |
|                                                                                  |             | MCWB   | 44.9  | 46.6  | 50.0  | 54.7  | 62.9  | 68.9  | 70.6  | 70.9  | 67.1  | 60.8  | 55.5  | 48.6  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 53.9  | 56.4  | 58.2  | 63.2  | 69.8  | 75.5  | 77.2  | 79.1  | 73.7  | 68.1  | 63.0  | 57.4  |
|                                                                                  |             | MCDB   | 59.1  | 63.1  | 68.9  | 74.0  | 79.3  | 84.3  | 84.3  | 85.4  | 79.9  | 74.8  | 70.3  | 62.7  |
|                                                                                  | 2%          | WB     | 50.3  | 53.2  | 55.1  | 59.9  | 67.6  | 73.2  | 75.4  | 76.3  | 71.3  | 65.5  | 60.6  | 54.4  |
|                                                                                  |             | MCDB   | 54.2  | 59.5  | 64.2  | 69.0  | 75.8  | 81.7  | 83.3  | 83.7  | 77.9  | 71.6  | 66.1  | 58.4  |
|                                                                                  | 5%          | WB     | 47.1  | 50.2  | 52.6  | 57.4  | 65.6  | 71.7  | 73.9  | 74.5  | 69.7  | 63.7  | 58.8  | 51.4  |
|                                                                                  |             | MCDB   | 49.4  | 55.2  | 60.2  | 65.1  | 72.9  | 79.7  | 82.0  | 82.3  | 76.3  | 69.2  | 63.3  | 54.1  |
|                                                                                  | 10%         | WB     | 45.0  | 47.1  | 50.3  | 55.6  | 63.9  | 70.1  | 72.5  | 73.0  | 68.2  | 61.8  | 56.3  | 48.9  |
|                                                                                  |             | MCDB   | 46.7  | 50.8  | 56.0  | 61.3  | 70.5  | 78.2  | 80.8  | 81.0  | 75.1  | 66.7  | 59.6  | 51.2  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 12.8  | 14.9  | 17.0  | 17.3  | 18.0  | 18.3  | 18.5  | 18.6  | 17.8  | 15.9  | 14.1  | 13.0  |
|                                                                                  |             | MCDBR  | 17.6  | 21.9  | 25.0  | 23.1  | 22.3  | 21.8  | 21.2  | 21.3  | 20.8  | 20.6  | 18.7  | 16.6  |
|                                                                                  |             | MCWBR  | 12.5  | 14.3  | 14.4  | 12.7  | 11.8  | 11.3  | 11.1  | 11.6  | 11.2  | 11.8  | 11.5  | 12.2  |
|                                                                                  | 5% WB       | MCDBR  | 15.8  | 21.0  | 22.9  | 21.4  | 20.0  | 19.6  | 19.1  | 19.6  | 18.8  | 18.5  | 17.0  | 14.9  |
|                                                                                  |             | MCWBR  | 11.9  | 14.2  | 13.7  | 12.2  | 11.2  | 11.0  | 10.8  | 11.4  | 10.9  | 11.4  | 10.5  | 11.6  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.363 | 0.395 | 0.434 | 0.482 | 0.451 | 0.452 | 0.459 | 0.468 | 0.447 | 0.413 | 0.386 | 0.366 |
|                                                                                  | taud        |        | 2.298 | 2.204 | 2.121 | 2.000 | 2.172 | 2.216 | 2.212 | 2.178 | 2.200 | 2.263 | 2.275 | 2.286 |
|                                                                                  | Ebn at Noon |        | 239   | 249   | 254   | 250   | 263   | 262   | 259   | 252   | 247   | 240   | 226   | 224   |
|                                                                                  | Edh at Noon |        | 28    | 36    | 44    | 54    | 46    | 44    | 44    | 44    | 40    | 34    | 29    | 26    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 467   | 692   | 1061  | 1500  | 1901  | 2098  | 2122  | 1905  | 1368  | 860   | 534   | 405   |
|                                                                                  | RadStd      |        | 61    | 82    | 102   | 151   | 128   | 155   | 128   | 94    | 119   | 94    | 71    | 60    |